

Lesson 10: Data Preparation and Transformation using Tableau Prep

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What you will learn ?

- a) Connect and load data into Tableau Prep
- b) Clean and profile data
- c) Group and standardize values
- d) Merge mismatched fields
- e) Perform aggregations in Prep
- f) Apply filters and transformations
- g) Create and automate data preparation flows

You need Tableau Prep Builder for this Lesson. You need to download and install. Here are the steps :

1. Download

- Download the Windows installer (.exe) directly from the official [Tableau Prep Builder Releases Page](#) (or use the [Tableau Prep Free Trial Page](#)).

2. Install

- Open your **Downloads** folder and run the .exe file.
- Accept the license agreement, click **Install**, and confirm **Yes** on the Windows prompt.

3. Activate & Launch

- Open **Tableau Prep Builder** from the Start Menu.
- Select **Activate with a product key** (enter your Tableau Desktop/Student key) or **Start a Free Trial**.

Dataset: Custom Designed Students Dataset [csv files]

You can also create database using Python code. Here is the code

```
import pandas as pd
import numpy as np

# We need custom CSV datasets specifically designed for Tableau
# Prep features in Practical 10:
# a) Connect & load data
# b) Clean and profile data (data type issues, extra spaces,
# nulls, outliers)
# c) Group and standardize values (pronunciation/spelling
# variations, e.g. "Comp Sci", "CS", "Computer Science", "CSE",
# "Electronic Eng", "ECE")
# d) Merge mismatched fields (e.g., across two unioned files
# or separate columns: "Maths" vs "Mathematics", "Std_ID" vs
# "Student_ID")
# e) Perform aggregations in Prep (e.g., Aggregate step: Total
# Score, Subject Average, Group by Student/Term)
# f) Apply filters and transformations (e.g., Filter out
# dropped students, calculate weighted GPA / Grade, split
# columns)
```

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```
# g) Create and automate data preparation flows (Output step
to .hyper / .csv)

# Table 1: Raw Student Assessment Log (Term 1) ->
Student_Assessments_Term1.csv
# Contains messy department names for (c) Group & Standardize,
inconsistent casings, text dates, extra spaces

term1_data = {
    'Student_ID': ['S101', 'S101', 'S102', 'S102', 'S103',
'S103', 'S104', 'S104', 'S105', 'S105', 'S106', 'S106',
'S107', 'S107', 'S108', 'S108'],
    'Student_Name': ['Aarav Sharma ', 'Aarav Sharma', 'priya
VERMA', 'priya VERMA', 'Rohan Gupta', 'Rohan Gupta', 'Sneha
Patel', 'Sneha Patel', 'aditya Rao ', 'aditya Rao ', 'Ananya
Singh', 'Ananya Singh', 'Rahul Nair', 'Rahul Nair', 'Kavya
Reddy', 'Kavya Reddy'],
    'Department': ['Computer Science', 'Comp Sci',
'Electronics', 'ECE', 'CS', 'CSE', 'Information Tech', 'IT',
'Mechanical', 'Mech Engg', 'Computer Science', 'CS', 'ECE',
'Electronics Engg', 'IT', 'Info Tech'],
    'Assessment_Type': ['Midterm', 'Final', 'Midterm',
'Final', 'Midterm', 'Final', 'Midterm', 'Final', 'Midterm',
'Final', 'Midterm', 'Final', 'Midterm', 'Final', 'Midterm',
'Final'],
    'Exam_Date': ['2025/09/15', '2025/11/20', '15-09-2025',
'20-11-2025', '2025.09.15', '2025.11.20', '2025/09/15',
'2025/11/20', '2025/09/15', '2025/11/20', '2025/09/15',
'2025/11/20', '2025/09/15', '2025/11/20', '2025/09/15',
'2025/11/20'],
    'Subject_Code': ['CS101', 'CS101', 'EC101', 'EC101',
'CS101', 'CS101', 'IT101', 'IT101', 'ME101', 'ME101', 'CS101',
'CS101', 'EC101', 'EC101', 'IT101', 'IT101'],
    'Marks_Obtained': [85, 92, 70, 78, 45, 52, 90, 96, 35, 40,
88, 94, 65, 72, 92, 98],
    'Max_Marks': [100, 100, 100, 100, 100, 100, 100, 100, 100,
100, 100, 100, 100, 100, 100, 100],
    'Enrollment_Status': ['Active', 'Active', 'Active',
'Active', 'Active', 'Active', 'Dropped',
'Dropped', 'Active', 'Active', 'Active', 'Active', 'Active',
'Active']
}

# Table 2: Student Assessment Log (Term 2) with Mismatched
Column Headers -> Student_Assessments_Term2.csv
# For (d) Merge Mismatched Fields (e.g. Std_ID instead of
Student_ID, Score instead of Marks_Obtained, Dept instead of
Department)

term2_data = {
    'Std_ID': ['S101', 'S101', 'S102', 'S102', 'S103', 'S103',
'S104', 'S104', 'S106', 'S106', 'S107', 'S107', 'S108',
'S108'],
```

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```

'Full_Name': ['Aarav Sharma', 'Aarav Sharma', 'Priya
Verma', 'Priya Verma', 'Rohan Gupta', 'Rohan Gupta', 'Sneha
Patel', 'Sneha Patel', 'Ananya Singh', 'Ananya Singh', 'Rahul
Nair', 'Rahul Nair', 'Kavya Reddy', 'Kavya Reddy'],
'Dept': ['Computer Science', 'CS', 'ECE', 'Electronics',
'CSE', 'CS', 'IT', 'Information Tech', 'CSE', 'Computer
Science', 'ECE', 'Electronics', 'IT', 'IT'],
'Exam_Type': ['Midterm', 'Final', 'Midterm', 'Final',
'Midterm', 'Final', 'Midterm', 'Final', 'Midterm', 'Final',
'Midterm', 'Final', 'Midterm', 'Final'],
'Assessment_Date': ['2026/02/10', '2026/04/25',
'2026/02/10', '2026/04/25', '2026/02/10', '2026/04/25',
'2026/02/10', '2026/04/25', '2026/02/10', '2026/04/25',
'2026/02/10', '2026/04/25', '2026/02/10', '2026/04/25'],
'Course_Code': ['CS201', 'CS201', 'EC201', 'EC201',
'CS201', 'CS201', 'IT201', 'IT201', 'CS201', 'CS201', 'EC201',
'EC201', 'IT201', 'IT201'],
'Score': [88, 95, 75, 82, 50, 60, 94, 98, 90, 96, 70, 75,
95, 100],
'Total_Possible': [100, 100, 100, 100, 100, 100, 100, 100, 100, 100,
100, 100, 100, 100],
'Enrollment_Status': ['Active', 'Active', 'Active',
'Active', 'Active', 'Active', 'Active', 'Active',
'Active', 'Active', 'Active', 'Active', 'Active']
}

# Table 3: Course Credit & Grading Benchmark ->
Course_Credit_Master.csv

course_data = {
    'Subject_Code': ['CS101', 'CS201', 'EC101', 'EC201',
'IT101', 'IT201', 'ME101'],
    'Subject_Title': ['Data Structures', 'Algorithms',
'Digital Logic', 'Signals & Systems', 'Database Systems',
'Cloud Computing', 'Thermodynamics'],
    'Credit_Hours': [4, 4, 3, 4, 4, 4, 3],
    'Passing_Marks': [40, 40, 40, 40, 40, 40, 40]
}

df_t1 = pd.DataFrame(term1_data)
df_t2 = pd.DataFrame(term2_data)
df_c = pd.DataFrame(course_data)

df_t1.to_csv('Student_Assessments_Term1.csv', index=False)
df_t2.to_csv('Student_Assessments_Term2.csv', index=False)
df_c.to_csv('Course_Credit_Master.csv', index=False)

print("All 3 CSV datasets for Lesson 10 created successfully.")

```

Here is the complete schema design and CSV datasets tailored specifically for Tableau Prep Builder to fulfill all activities of Practical 10 (a through g).

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Dataset Architecture and Schema Design

To practice every feature of Tableau Prep (profiling, grouping, merging mismatched union columns, aggregations, calculated filters, and automated flow generation), we use 3 specialized CSV files:

1. Student_Assessments_Term1.csv (Term 1 Assessment Log)
 - Purpose: Data profiling, handling extra spaces, mixed casings, inconsistent date text strings, and spelling variations in departments.
 - Columns: Student_ID, Student_Name, Department, Assessment_Type, Exam_Date, Subject_Code, Marks_Obtained, Max_Marks, Enrollment_Status.
2. Student_Assessments_Term2.csv (Term 2 Assessment Log with Mismatched Schemas)
 - Purpose: Practicing Union and resolving mismatched field names (e.g., Std_ID vs Student_ID, Score vs Marks_Obtained, Course_Code vs Subject_Code).
 - Columns: Std_ID, Full_Name, Dept, Exam_Type, Assessment_Date, Course_Code, Score, Total_Possible, Enrollment_Status.
3. Course_Credit_Master.csv (Subject Metadata & Weightages)
 - Purpose: Prep Joins, weighted GPA calculations, credit hours, and subject categorization.
 - Columns: Subject_Code, Subject_Title, Credit_Hours, Passing_Marks.

Mapping to Practical 10 Objectives

Sub-task	Activity	Practical Execution in Tableau Prep
a	Connect and load data	Connect to text files (Term1, Term2, and Course_Master) in Tableau Prep.
b	Clean and profile data	Use Prep's Profile pane to inspect value distributions, summary cards, and detect data type errors.
c	Group and standardize values	Use Group Values (Pronunciation, Common Characters, or Manual) to unify "Comp Sci", "CS", "CSE" --> "Computer Science".
d	Merge mismatched fields	Union Term 1 and Term 2, then drag and drop mismatched columns (Std_ID onto Student_ID, Score onto Marks_Obtained) to merge them.
e	Perform aggregations in Prep	Add an Aggregate Step to compute student total score, overall average, and exam count grouped by Student_ID & Term.

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Sub-task	Activity	Practical Execution in Tableau Prep
f	Apply filters and transformations	Filter out "Dropped" enrollment records, create calculated fields for Percentage, Grade, and Status flags.
g	Create and automate data flows	Build an Output Step to export the cleaned, unified pipeline to a .hyper extract or .csv file.

Raw CSV Data (Create these 3 files on your computer). You can create different csv files using Notepad/text Editor. Copy paste the data in Notepad/text Editor and save the files with .csv extension.

File 1: Student_Assessments_Term1.csv

```
Student_ID,Student_Name,Department,Assessment_Type,Exam_Date,
Subject_Code,Marks_Obtained,Max_Marks,Enrollment_Status
S101, Aarav Sharma ,Computer
Science,Midterm,2025/09/15,CS101,85,100,Active
S101,Aarav Sharma,Comp
Sci,Final,2025/11/20,CS101,92,100,Active
S102,priya VERMA,Electronics,Midterm,15-09-
2025,EC101,70,100,Active
S102,priya VERMA,ECE,Final,20-11-2025,EC101,78,100,Active
S103,Rohan Gupta,CS,Midterm,2025.09.15,CS101,45,100,Active
S103,Rohan Gupta,CSE,Final,2025.11.20,CS101,52,100,Active
S104,Sneha Patel,Information
Tech,Midterm,2025/09/15,IT101,90,100,Active
S104,Sneha Patel,IT,Final,2025/11/20,IT101,96,100,Active
S105,aditya Rao
,Mechanical,Midterm,2025/09/15,ME101,35,100,Dropped
S105,aditya Rao ,Mech
Engg,Final,2025/11/20,ME101,40,100,Dropped
S106,Ananya Singh,Computer
Science,Midterm,2025/09/15,CS101,88,100,Active
S106,Ananya Singh,CS,Final,2025/11/20,CS101,94,100,Active
S107,Rahul Nair,ECE,Midterm,2025/09/15,EC101,65,100,Active
S107,Rahul Nair,Electronics
Engg,Final,2025/11/20,EC101,72,100,Active
S108,Kavya Reddy,IT,Midterm,2025/09/15,IT101,92,100,Active
S108,Kavya Reddy,Info
Tech,Final,2025/11/20,IT101,98,100,Active
```

File 2: Student_Assessments_Term2.csv

```
Std_ID,Full_Name,Dept,Exam_Type,Assessment_Date,Course_Code,S
core>Total_Possible,Enrollment_Status
S101,Aarav Sharma,Computer
Science,Midterm,2026/02/10,CS201,88,100,Active
S101,Aarav Sharma,CS,Final,2026/04/25,CS201,95,100,Active
S102,Priya Verma,ECE,Midterm,2026/02/10,EC201,75,100,Active
```

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```
S102,Priya
Verma,Electronics,Final,2026/04/25,EC201,82,100,Active
S103,Rohan Gupta,CSE,Midterm,2026/02/10,CS201,50,100,Active
S103,Rohan Gupta,CS,Final,2026/04/25,CS201,60,100,Active
S104,Sneha Patel,IT,Midterm,2026/02/10,IT201,94,100,Active
S104,Sneha Patel,Information
Tech,Final,2026/04/25,IT201,98,100,Active
S106,Ananya Singh,CSE,Midterm,2026/02/10,CS201,90,100,Active
S106,Ananya Singh,Computer
Science,Final,2026/04/25,CS201,96,100,Active
S107,Rahul Nair,ECE,Midterm,2026/02/10,EC201,70,100,Active
S107,Rahul
Nair,Electronics,Final,2026/04/25,EC201,75,100,Active
S108,Kavya Reddy,IT,Midterm,2026/02/10,IT201,95,100,Active
S108,Kavya Reddy,IT,Final,2026/04/25,IT201,100,100,Active
```

File 3: Course_Credit_Master.csv

```
Subject_Code,Subject_Title,Credit_Hours,Passing_Marks
CS101,Data Structures,4,40
CS201,Algorithms,4,40
EC101,Digital Logic,3,40
EC201,Signals & Systems,4,40
IT101,Database Systems,4,40
IT201,Cloud Computing,4,40
ME101,Thermodynamics,3,40
```

G: Create and Automate Data Preparation Flows

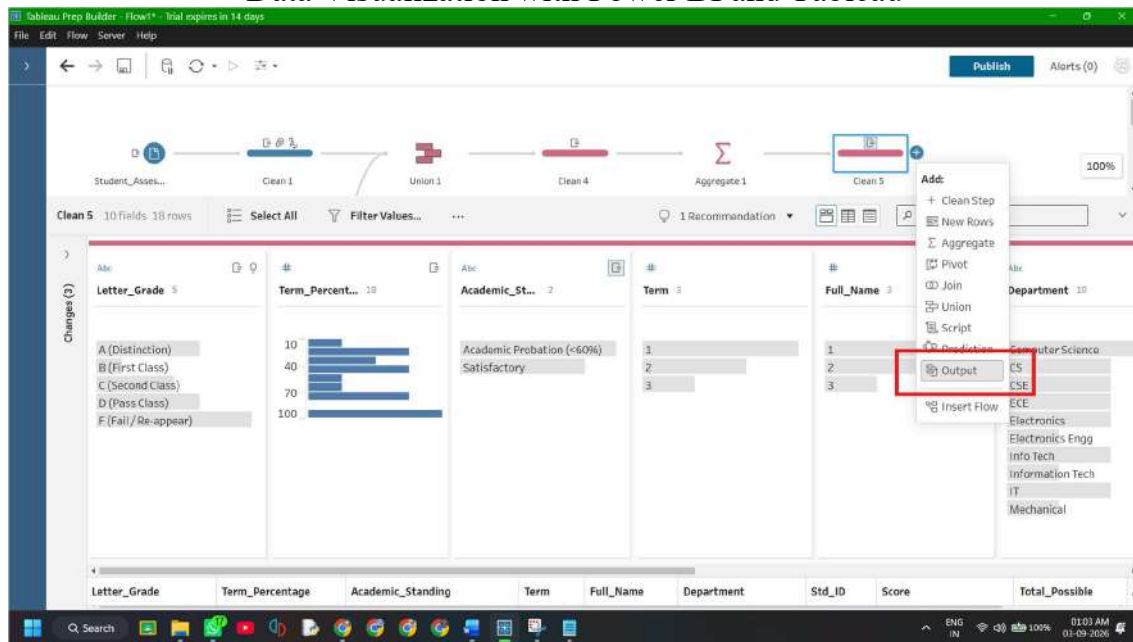
Objective: Configure an **Output Step** in Tableau Prep Builder to publish the fully transformed pipeline to an extract file (.hyper or .csv), and learn how to automate or execute the flow on demand.

Data preparation workflows must be repeatable and production-ready. Configuring a dedicated output node ensures that whenever new assessment scores arrive, running the flow automatically executes all trimming, grouping, merging, filtering, and scoring calculations in seconds without any manual rework.

Step 1: Add an Output Step to the End of the Pipeline

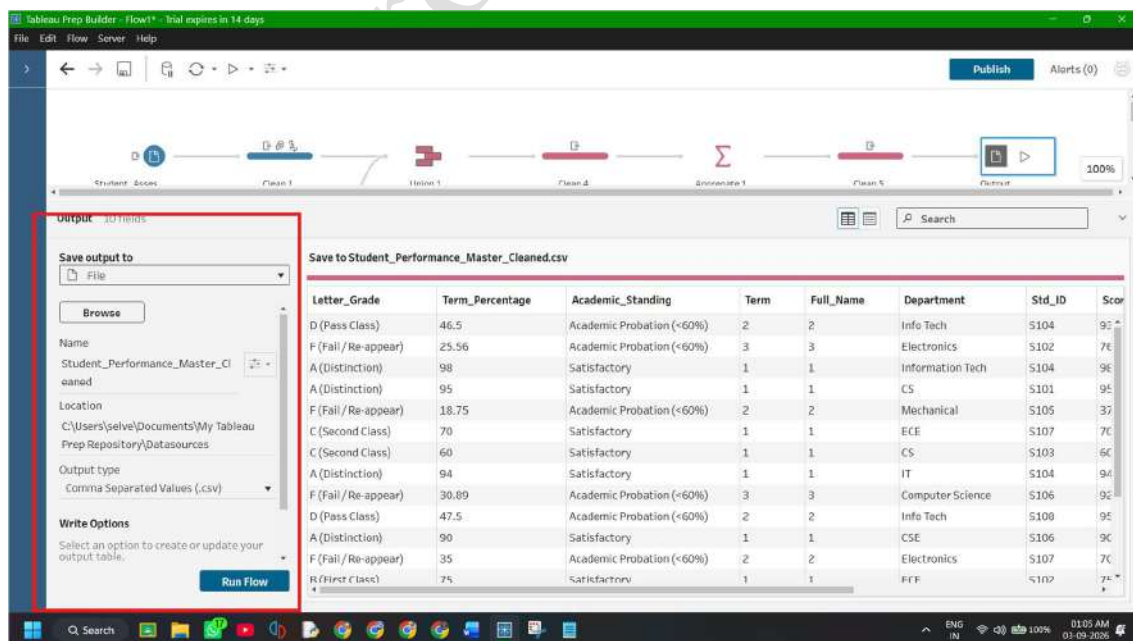
1. On the Flow canvas, locate the final cleaning step (**Clean 4**).
2. Hover over the **Clean 4** icon and click the + icon.
3. Select **Output** (represented by the download/save arrow icon).
4. An **Output** step node will appear connected directly to **Clean 4**.

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Step 2: Configure Output Settings in the Bottom Pane

- Click on the **Output** node to open its configuration panel below:
 - Save output to:** Select **File** (or *Tableau Server / Tableau Cloud*).
 - Output type:** Select **Tableau Data Extract (.hyper)** for high-speed dashboarding, or **Comma Delimited Text (.csv)** for universal export.
 - Location:** Click **Browse** and choose the directory where you want the final dataset saved.
 - File Name:** Enter *Student_Performance_Master_Cleaned*.
 - Write Option:** Select **Create table** (or **Append to table** if updating existing records).

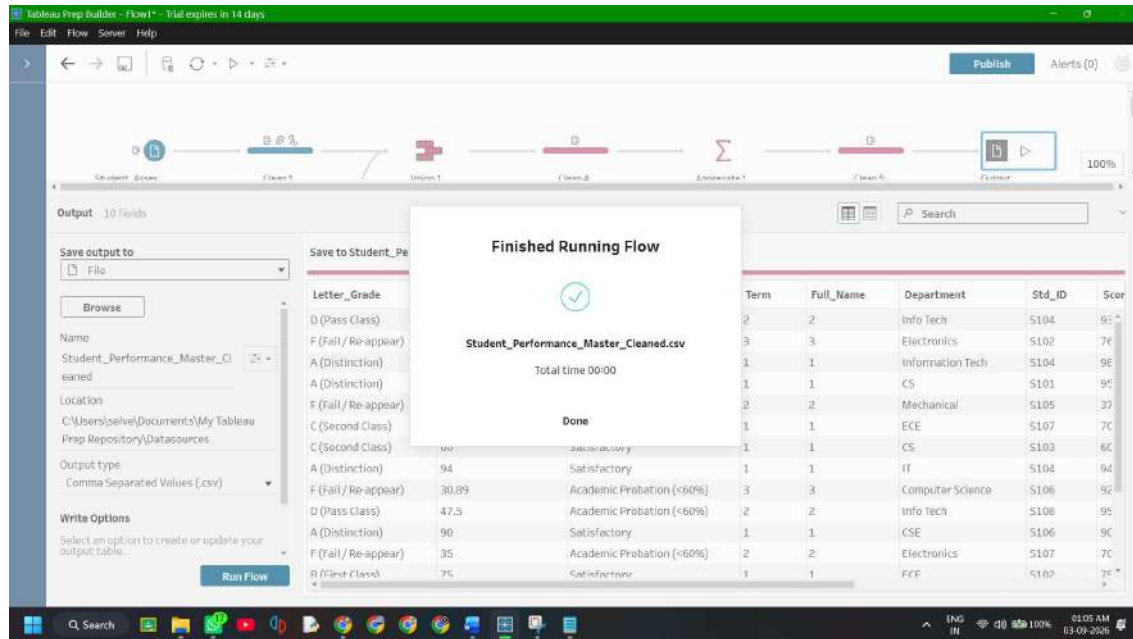


Step 3: Run the Data Prep Flow

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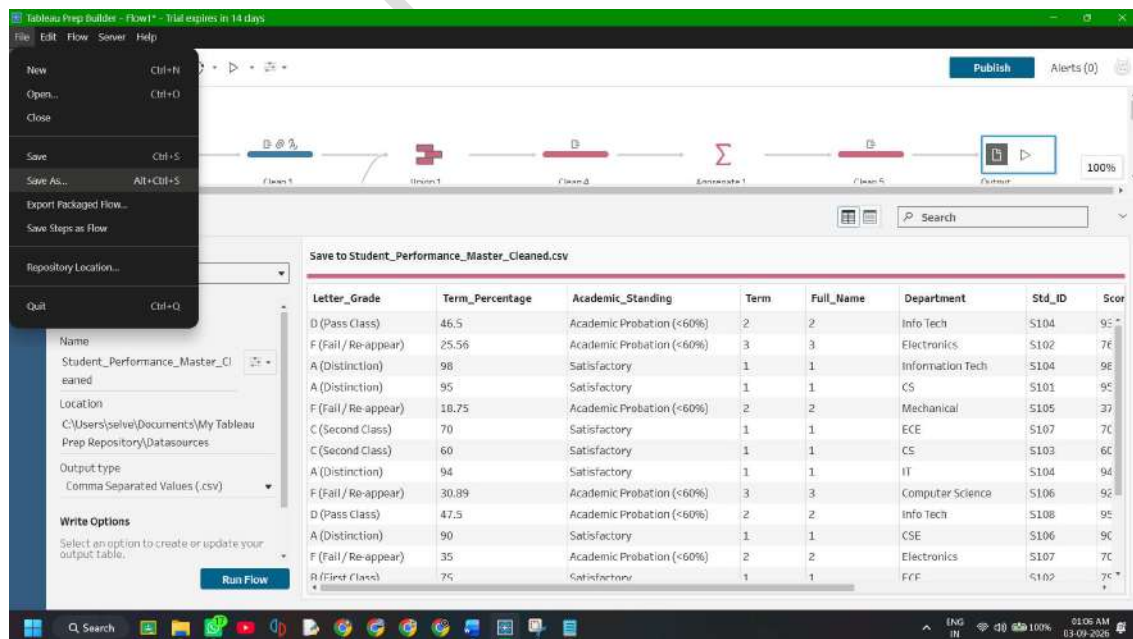
1. In the Output pane at the bottom-left, click the blue **Run Flow** button (or click the play icon $\blacktriangleright$ directly on the Output node on the canvas).
2. Tableau Prep will process all upstream nodes (Cleaning $\rightarrow$ Grouping $\rightarrow$ Unioning $\rightarrow$ Merging $\rightarrow$ Aggregating $\rightarrow$ Scoring).
3. A success notification will appear stating:

"The flow ran successfully in X seconds."



Step 4: Save and Automate the Flow File

1. Go to the top menu bar $\rightarrow$ select **File** $\rightarrow$ **Save As...**

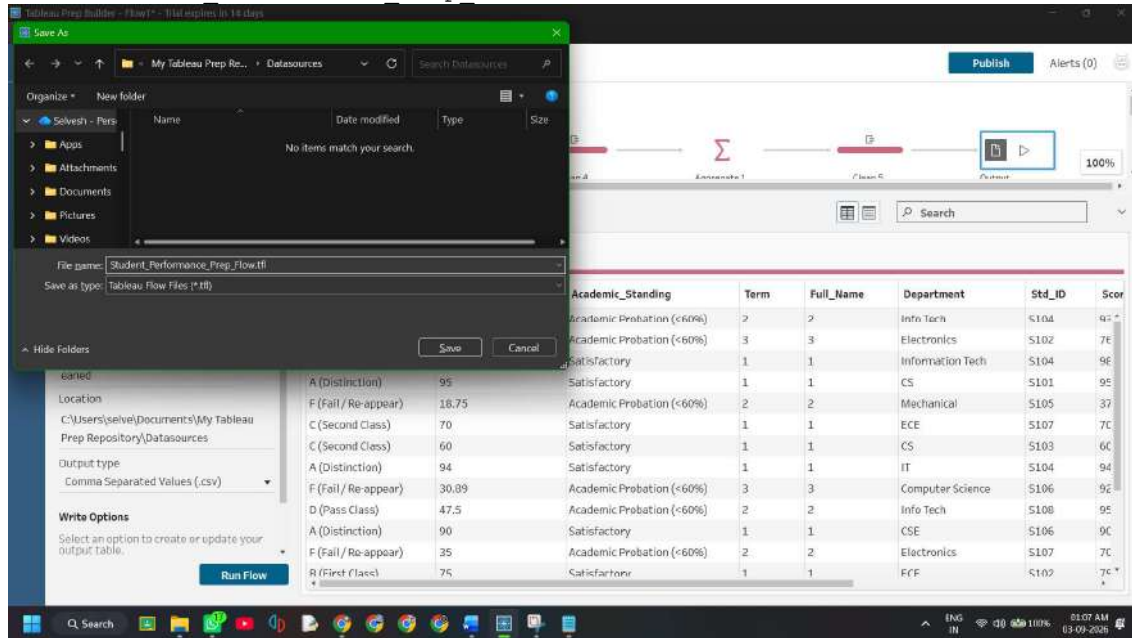


2. Name your workflow:

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Plaintext

Student Performance Prep Flow.tfl



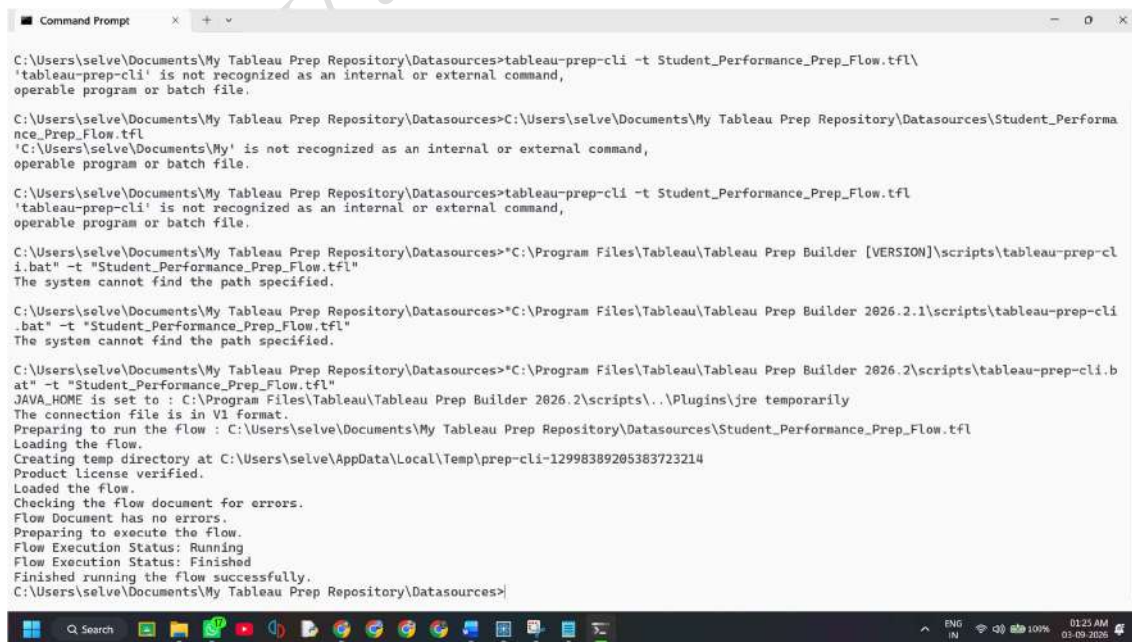
3. (Automation Options in Production):

- **Command Line Execution:** You can automate this flow on Windows/Mac using the Tableau Prep Command Line Interface (CLI):

Bash

```
tableau-prep-cli -t Student_Performance_Prep_Flow.tfl
```

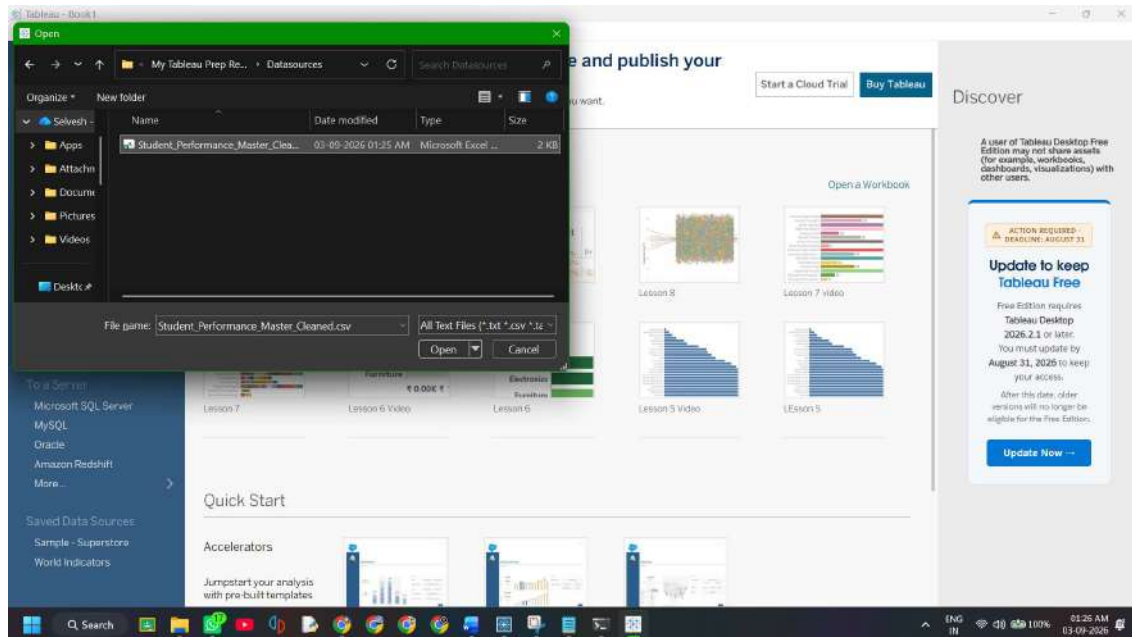
- **Tableau Prep Conductor:** If connected to Tableau Cloud/Server, you can publish the flow and set a scheduled refresh (e.g., every Monday at 6:00 AM) to automatically run whenever new semester files are added.



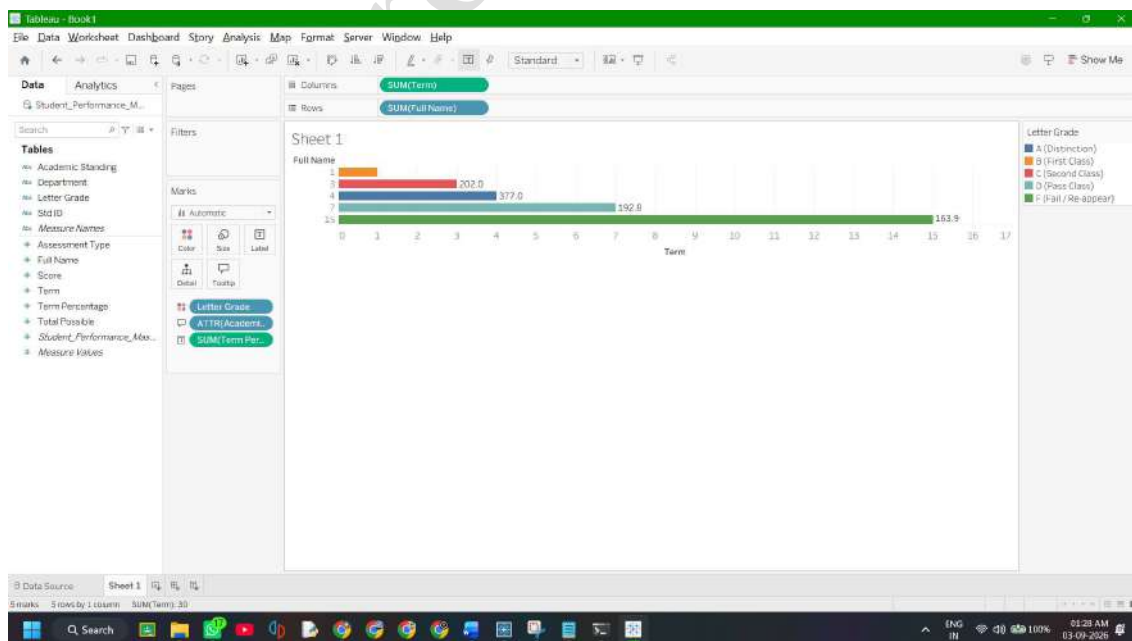
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Step 5: Verify the Final Master Dataset in Tableau Desktop

1. Open **Tableau Desktop**.
2. Connect to the exported **Student_Performance_Master_Cleaned.csv** (or .hyper).



3. Build the self-performance dashboard using the pre-calculated metrics:
 - Drag **Student_Name** to **Rows**.
 - Drag **Term** to **Columns**.
 - Drag **Term_Percentage** to **Text**.
 - Drag **Letter_Grade** to **Color**.
 - Drag **Academic_Standing** to **Tooltip**.

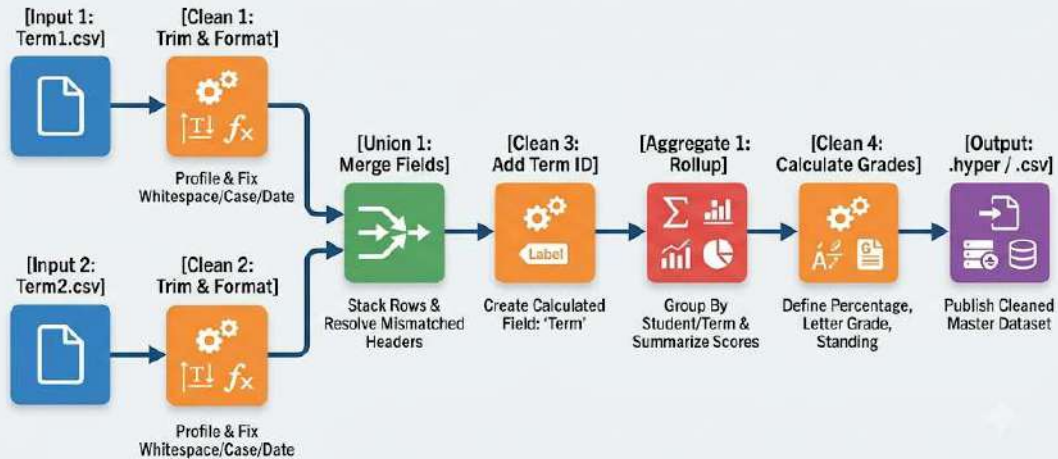


(In my Desktop Version of Tableau the Full_name[Student_Name] is showing numbers even after Changing values there will be different outcomes in your point of view)

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Complete End-to-End Flow Pipeline Summary

DATA PREPARATION & TRANSFORMATION FLOW (TABLEAU PREP)



Pipeline Stage	Step Type	Key Actions Performed	Output Result
Input Layer	Data Ingestion	Connect to Term1.csv, Term2.csv, and Course_Master.csv	Raw tables loaded on canvas
Cleaning Layer	Clean Steps 1 & 2	Trim whitespace, standardize dates, uppercase names	Structured single-source tables
Standardization	Group Values	Consolidate aliases (CS, CSE, Comp Sci → Computer Science)	Clean department dimensions
Integration Layer	Union Step	Stack Term 1 & 2; drag-and-drop mismatched headers	Unified schema (30

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Pipeline Stage	Step Type	Key Actions Performed	Output Result
			assessment rows)
Aggregation Layer	Aggregate Step	Group by student & term; compute Sum (Marks) and Avg (Score)	14 summary records
Business Logic	Clean Step 4	Filter dropped students; calculate %, Letter_Grade, and Standing	Analytical scorecard metrics
Export Layer	Output Step	Export to Student_Performance_Master_Cleaned.hyper / .csv	Production-ready dashboard source

~~~~~ End of Lesson 10 ~~~~~